

I1 implements C1

Program Code:

```
package core_java;

interface I1 {
    void bookTicket();
}

class C1 implements I1 {

    public void bookTicket() {
        System.out.println("Ticket booked successfully");
    }

    void showSeat() {
        System.out.println("Seat Number: A12");
    }

    void cancelTicket() {
        System.out.println("Ticket cancellation completed");
    }
}

public class I1toC1 {
    public static void main(String[] args) {

        C1 obj = new C1();

        obj.bookTicket();
        obj.showSeat();
        obj.cancelTicket();
    }
}
```

Explanation:

Java does not allow an interface to implement a class. Therefore, this program uses the valid Java relationship in which class C1 implements interface I1. The interface declares bookTicket(), and class C1 provides its implementation. Class C1 also contains the showSeat() and cancelTicket() methods.

Compilation and Execution Commands:

Save the file as: I1toC1.java

Compile:
javac -d . I1toC1.java

Run:
java core_java.I1toC1

Output Screenshot:



```
Ticket booked successfully

Seat Number: A12

Ticket cancellation completed
```

Verification:

The output confirms that class C1 successfully implements the method declared in interface I1 and executes its own additional methods.

Result:

Thus, the program was successfully executed and the required interface and class relationship was demonstrated using valid Java syntax.